AI-Powered Smart Hydration & Wellness Wristband

05th June 2026

Before we finalize the components and go in to development, I had some quick questions

1. Accelerometer:

- You mentioned step count and sleep detection once which require Accelerometer but you did not mention about that in your detailed requirements for hardware prototype and Firmware. Just wanted to confirm we need to add an Accelerometer and include step and sleep detection in the Firmware.
- I have already researched about what accelerometer we should use just getting a heads up as I need a complete amount of peripherals requirement to get a good estimate of battery management etc.

2. AI implementation:

- Just to make sure I understood the requirements right, from what I understand AI how ever it is implemented is done on the App side not on the MCU/hardware side. As there is no need and also no meaningful way of integration for it on the hardware side.

3. Data logging in case of no BLE connection:

- You mentioned that the band should work independently even in the absence of a BLE connected device or without the app. This works logically for the haptic feedbacks but what about the data logging? We can do a few things:
    - We can store the Spo2, heart rate and step counts in the MCU flash memory. But obviously that flash memory has a limit and depends on for how long are we disconnected from the App.
    - Just store step counts in the flash memory as it is way easy and less space consuming for the Flash memory. Also, I think Spo2 and HR are only meaningful when you can view them in real time unless you want the complete days history (which according to your battery requirement you do not even intend it to be worn all day long).
- It all depends on whether we want the sensors to continuously read data like HR, Spo2 and Step count etc. Even in the absence of a device, it can stream that to? Also, you mentioned implementation of sleep modes which seems unlikely as we

are reading from sensors constantly. But we can still use an inherently low power MCU something from Nrf and have good battery performance even while constantly reading sensor data.